TEK306- Assignment for Week40- Solution

1. What are the advantages and disadvantages of hacking?

Advantages	Disadvantages
It can be used to prevent security attacks	It creates massive security issues
To plug the bugs and loopholes	Get unauthorized system access
It helps to prevent data theft	Stealing private information
Hacking prevents malicious attacks	Violating privacy regulations

2. What can an ethical hacker do?

An ethical hacker is a computer system and networking master who systematically endeavors to infiltrate a PC framework or network for the benefit of its owners to find security vulnerabilities that a malicious hacker could potentially exploit.

3. What is network sniffing?

System sniffing includes utilizing sniffer tools that empower real- time monitoring and analysis of data streaming over PC systems. Sniffers can be utilized for various purposes, regardless of whether it's to steal data or manage systems. Network sniffing is utilized for ethical and unethical purposes. System administrators utilize these as system monitoring and analysis tool to analyze and avoid network related issues, for example, traffic bottlenecks. Cyber criminals utilize these devices for untrustworthy purposes, for example, character usurpation, email, delicate information hijacking etc.

4. What is a firewall?

A firewall could be a device that allows/blocks traffic as per outlined set of rules. These are placed on the boundary of trusted and untrusted networks.

5. What are the tools used for ethical hacking?

There are several moral hacking tools out there within the marketing for different purposes, they are:

- NMAP – NMAP stands for Network Mapper- an open source tool that's used wide for network discovery and security auditing.
- Metasploit – Metasploit is one amongst the most powerful exploit tool to conduct basic penetration tests.
- Burp Suit – Burp Suite could be a widespread platform that's widely used for playing security testing of internet applications.
- Angry IP Scanner – Angry information processing scanner could be a lightweight, cross-platform information processing address and port scanner.
- Ettercap – Ettercap stands for local area network Capture. It is used for Man-in-the-Middle attack using a network security tool.

6. Explain how you can stop your website getting hacked?

By adapting the following methodology you'll be able to stop your website from being hacked:

- Using Firewall: Firewall can drop traffic from suspicious information processing and address if attack is an easy DOS
- Encrypting the Cookies: Cookie or Session poisoning may be prevented by encrypting the content of the cookies, associating cookies with the consumer information processing and temporal arrangement out the cookies once it slow
- Validating and confirmative user input: This approach is prepared to stop the type tempering by confirmative and verifying the user input before processing it
- Header Sanitizing and validation: This technique is beneficial against cross website scripting or XSS, this method includes verifying and sanitizing headers, parameters passed via the address, type parameters and hidden values to cut back XSS attacks.

7. What is Burp Suite? What are the tools does it contain?

Burp Suite is an integrated platform for performing security testing of web applications. It contains all the tools a hacker would need for attacking any application. A number of these functionalities are: Proxy, Spider, Scanner, Intruder, Repeater, Decoder, Comparer, and Sequencer.

8. What is SQL injection and its types?

If the application doesn’t sanitize the user input, then the SQL injection happens. Thus, a malicious hacker would inject SQL question to gain unauthorized access and execute administration operations on the database. SQL injections may be classified as- Error-based SQL injection, Blind SQL injection, Time-based SQL injection, etc.